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SCISSOR MECHANISM CLAMPING ASSEMBLY FOR A FILTER COMPARTMENT

Abstract

Implementations disclosed herein provide a compact and easy to maintain downdraft table for use in facilities with limited floor space that includes a unique filter compartment and exhaust fan assembly located beneath a perforated work surface. The filter compartment may include at least one rectangular filter insert that is removably held in place using a scissor mechanism clamping assembly. The scissor mechanism clamping assembly allows an operator to easily apply a distributed force on the rectangular filter insert to press the filter insert against a filter seat and lock the filter insert in that position. The scissor mechanism clamping assembly further allows the operator to easily release the distributed force from the rectangular filter insert to unseat the filter insert from the filter seat and remove the filter insert from the filter compartment for cleaning or replacement.

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Background/Summary

BACKGROUND

[0001] Exposure to poor air in industrial facilities can affect the health and safety of employees. Downdraft tables are used to collect dust, fumes, smoke, and particulates during certain industrial operations and applications. Downdraft tables can provide air filtration and maintain the air quality and control in a work environment.

SUMMARY

[0002] Implementations disclosed herein provide a downdraft table comprising a perforated work surface and a clamping assembly. The clamping assembly includes at least two scissor mechanisms, each mounted on opposing sides of a filter intake. One end of each of the scissor mechanisms mounted to a fixed position within the filter intake. The clamping assembly further includes two translating bars, an opposite end of each of the scissor mechanisms mounted to one of the translating bars. The clamping assembly still further includes an adjustment screw to change the position of folding legs making up each of the scissor mechanisms and thereby change the position of the translating bars. The downdraft table also includes a filter cavity to receive a filter insert to be pressed against a filter seat at a fan plenum by the translating pressure bars. The downdraft table further includes a fan assembly to draw air through the perforated work surface, the clamping assembly, the filter insert, and the fan plenum and exhaust filtered air out of the downdraft table.

[0003] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Other features, details, utilities, and advantages of the claimed subject matter will be apparent from the following more particular written Detailed Description of various implementations as further illustrated in the accompanying drawings and defined in the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. 1 illustrates a perspective exterior view of an example downdraft table that incorporates a scissor mechanism clamping assembly according for an associated filter compartment.

[0005] FIG. 2 illustrates a sectional diagrammatic interior view of an example downdraft table that incorporates a scissor mechanism clamping assembly for an associated filter compartment.

[0006] FIG. 3 illustrates an exploded view of an example filter compartment including a scissor mechanism clamping assembly and an exhaust fan assembly.

[0007] FIG. 4 illustrates a perspective view of an example filter compartment including a scissor mechanism clamping assembly.

[0008] FIG. 5 illustrates a perspective view of an example scissor mechanism clamping assembly for a filter compartment.

[0009] FIG. 6 illustrates a perspective view of an example scissor mechanism for a scissor mechanism clamping assembly.

[0010] FIG. 7A illustrates a schematic view of a scissor mechanism clamping assembly for a filter compartment in a retracted orientation.

[0011] FIG. 7B illustrates a schematic view of the scissor mechanism clamping assembly of FIG. 7A in an extended orientation.

[0012] FIG. 8 illustrates a flowchart of example air filtration operations in an example downdraft

table.

DETAILED DESCRIPTION

[0013] The disclosed technology includes a downdraft table for industrial applications that require air filtration. Specifically, the downdraft table may include a compact, easy to maintain downdraft table for use in facilities with limited floor space that includes a unique filter compartment and exhaust fan assembly located beneath the work surface. The filter compartment may include at least one rectangular filter insert that is removably held in place using a scissor mechanism clamping assembly. The scissor mechanism clamping assembly allows an operator to easily apply a distributed force on the rectangular filter insert to press the filter insert against a filter seat and lock the filter insert in that position. The scissor mechanism clamping assembly further allows the operator to easily release the distributed force from the rectangular filter insert to unseat the filter insert from the filter seat and remove the filter insert from the filter compartment. Dirty air and clean air are terms used herein to reference the state of an air flow upstream of the filter insert (i.e., dirty air) and downstream of the filter insert (i.e., clean air). Dirty air and clean air are not used herein to define an actual state of cleanliness of the air, as that may vary widely in application. In various implementations, the downdraft table may also include a spark arrestor for fire protection, one or more removable clean-out trays for particulate collection, and a louvre for one-way airflow, for example. All of the aforementioned components may be located in a main housing of the downdraft table.

[0014] FIG. 1 illustrates a perspective exterior view of an example downdraft table **100**. The downdraft table **100** is generally used to collect ambient air in the vicinity of the downdraft table **100** and filter that air before it is discharged back into the atmosphere. This can vastly improve air quality in the vicinity of the downdraft table **100**, particularly for operators that are working in the vicinity of the downdraft table **100**, including operators that are using the downdraft table **100** as a work surface. In some implementations, the downdraft table **100** includes caster wheels (e.g., wheel **112**) that permit operators to move and position the downdraft table **100** as desired. In other implementations, the disclosed technology may be included in an air filtration system independent of the downdraft table **100**.

[0015] The downdraft table **100** functions by pulling air through a perforated work surface **102** and/or a perforated rear wall **104**. Thus, air in FIG. 1 can be pulled downward and/or rearward through the perforated holes into the downdraft table **100** by a fan (not shown, see e.g., fan **248** of FIG. 2), which is located below the perforated work surface **102**. The perforations may be consistently spaced (e.g., as illustrated on the perforated work surface **102**) or have some variation in pattern (e.g., as illustrated on the perforated rear wall **104**). In other implementations, for example in an air filtration system, air can be pulled through an intake, panel, nozzle, or other entry way aside from the perforated work surface **102** and/or the perforated rear wall **104**.

[0016] As noted above, the downdraft table **100** has one or both of downdraft and backdraft airflow, via the perforations or perforated holes in the perforated work surface **102** and/or the perforated rear wall **104**. The perforations can vary in size and shape, for example, the perforations can be $\frac{1}{2}'' \times \frac{3}{16}''$ or $\frac{1}{2}'' \times \frac{3}{8}''$ rectangular holes in 16GA thick material. For further example, the perforations can be $\frac{3}{16}''$, $\frac{3}{8}''$, or $\frac{1}{2}''$ diameter circular holes in $\frac{1}{2}''$ thick material. In some implementations, the perforations are arranged on multiple grates that make up the perforated work surface **102** and/or the perforated rear wall **104**. For example, the perforations may be configured into removable grates, such as removable grate **118** (illustrated by an area of uniformly distributed perforations bounded by parallel broken lines). These grates may be 5" wide, 1" thick, and span the length of the work surface (e.g., 36", 48", 60", 72", etc. long), for example. Other work surface configurations are contemplated. The vacuum of the fan in the downdraft table **100** provides even dispersion of negative air pressure and a resulting inward airflow over the perforated work surface **102** and/or the perforated rear wall **104** of the downdraft table **100**.

[0017] The downdraft table **100** will have a power supply appropriate for power needs of the

downdraft table **100**, including but not limited to a motor driving the fan and various sensors and controls for the downdraft table **100**. The downdraft table **100** may be configured for single phase (115V/230V) or three phase (208-230V/460V) power, for example. The controls (e.g., an on/off switch **108**, a pressure gauge **106**, etc.) are located on a front panel **152** of the downdraft table **100**. In other implementations, the controls may be located in different areas of the downdraft table **100**. The on/off switch **108** for a motor and its associated fan selectively pulls air from the perforated work surface **102** and/or the perforated rear wall **104** down into the downdraft table **100** and out through an exhaust (not shown) in the back of the downdraft table **100**. In some implementations, the on/off switch **108** may be a manual toggle switch that acts as motor starter. The controls may be housed in a NEMA 12/4X enclosure and contain thermal protection that protects an operator from electrical issues.

[0018] As vacuum builds up in the downdraft table **100**, the pressure gauge **106** measures the air pressure and indicates to the operator if and when the vacuum meets a predetermined differential pressure threshold. In some implementations, the predetermined differential pressure threshold can be a static reading. The pressure gauge **106** provides the operator with an indication of the cleanliness of an internal filter insert **110** (depicted in broken lines as it is hidden from view by access door **116**), as a dirty filter insert will cause a vacuum pressure to exceed the differential pressure threshold. Through this feedback, the operator will decide based on the indication of the pressure gauge **106** of when to clean or replace the filter insert **110**.

[0019] As air, including suspended particulates, is pulled from the exterior of the downdraft table **100**, it moves downward and towards the left interior side of the downdraft table **100**, for example. This example general airflow direction is depicted in FIG. 2. In other implementations, airflow can be moved in other general directions (e.g., right, front, or back). In some implementations, the air moves through a spark arrestor (not shown, see e.g., spark arrestor **226** of FIG. 2) located under the perforated work surface **102** on the left interior side of the downdraft table **100** between the filter insert **110** and the perforated work surface **102**. The spark arrestor adds protection against applications that may create sparks or increased risk of a fire such as welding, grinding metal, or plasma cutting. The air moves through the spark arrestor downward in the downdraft table **100**, and then moves in a direction toward the center of the downdraft table **100** into a filter compartment (not shown, see e.g., filter compartment **224** of FIG. 2) that houses the filter insert **110**.

[0020] As the air moves through the filter insert **110**, particulates suspended within the air fall out of the airflow via gravity into one or more removable clean-out trays (not shown, see e.g., trays **212**, **214** of FIG. 2) or attach to filter media within the filter insert **110** prior to being ingested into the fan and exhausted out of an exhaust port (not shown, see e.g., exhaust port **222** of FIG. 2) located on a side, rear, or bottom panel of the downdraft table **100**. In various implementations, the exhaust port may take the form of a perforated area, area of expanded metal, or a vent pipe on any outside-facing panel of the downdraft table **100**. In some implementations, particulate is precluded from blow-back towards the operator by a one-way self-closing louvre (not shown, see e.g., louvre **230** of FIG. 2) that allows dirty air to move through the downdraft table **100** in only one direction, thus preventing reverse airflow through the perforated work surface **102** and/or the perforated rear wall **104** and protecting the operator from potential particulate blow-back.

[0021] The filter insert **110** in the downdraft table **100** is designed to address applications that produce dust or fumes on a continuous basis. The operator can observe the pressure gauge **106** to help determine the cleanliness of the filter insert **110**. The pressure gauge **106** primarily measures the differential pressure, or pressure drop, across the filter insert **110**. When the filter insert **110** has met the predetermined differential pressure threshold and needs to be cleaned or replaced, a notification system can send a signal to alert the operator. The threshold can be a specific pressure range indicating sufficient particulates have accumulated on the filter insert **110** to restrict airflow.

[0022] The access door **116** depicted on the front panel **152** of the downdraft table **100** provides access to the filter compartment, including the filter insert **110**, a scissor mechanism clamping

assembly **120** (depicted in broken lines as it is hidden from view by the access door **116**) for securing the filter insert **110** within the filter compartment, and the removable clean-out trays. The downdraft table **100** may have different or additional access doors to that depicted in FIG. **1**. For example, another door may be located on the non-depicted side of the downdraft table **100** for access to a right-side interior of the downdraft table **100** for maintenance (e.g., access to the motor, fan, etc.).

[0023] The filter insert **110** is removable and replaceable by hand by manipulating the scissor mechanism clamping assembly **120**. More specifically, the filter insert **110** is held in place by the scissor mechanism clamping assembly **120** that applies a uniform force on a perimeter of the filter insert **110** to press and secure the filter insert **110** against a filter seat (not shown, see e.g., filter seat **264** of FIG. **2**). The scissor mechanism clamping assembly **120** functions by transforming rotational motion applied to a scissor mechanism handle (not shown, see e.g., scissor mechanism handle **582** of FIG. **5**) in a first direction by an operator into a translational motion that seats the filter insert **110** to a fan plenum (not shown, see e.g., fan plenum **232**). Rotational motion applied to the scissor mechanism handle in a second opposite direction by the operator drives an opposite translational motion that unseats the filter insert **110** from the fan plenum so that the filter insert **110** may be removed from the filter compartment. In some implementations, a filter gasket may form part of the filter seat and may be compressed between the filter insert **110** and the fan plenum to create the desired seal. As a result, changing filter inserts requires no tools and access to the filter compartment requires only the straightforward removal of the access door **116**.

[0024] The downdraft table can be a modular design, and tailored to desired applications and sizes, as well as operator requirements. For example, the following downdraft tables with the perforated work surface **102** and the perforated rear wall **104** can be configured for approximately 2000 CFM, 30"D×48"W×58"H; 2500 CFM, 30"D×60"W×58"H; 3000CFM, 30"D×72"W×58"H; 4000 CFM, 30"D×96"W×58"H; 2000 CFM, 30"D×48"W×80"H; 2500 CFM, 30"D×48"W×80"H; 3000 CFM, 30"D×48"W×80"H; or 4000 CFM, 30"D×48"W×80"H. For further example, the following flat top downdraft tables with the perforated work surface **102** only can be configured for approximately 2000 CFM, 30"D×48"W×34"H; 2500 CFM, 30"D×48"W×34"H; 3000 CFM, 30"D×48"W×34"H; or 4000 CFM, 30"D×48"W×34"H. Still further, depth of the downdraft tables may be customized using regular increments, such as 5" (e.g., 35", 40", 45", 50", 55", and 60"). The forgoing are merely examples as other configurations and sizes of the downdraft table **100** are contemplated herein.

[0025] FIG. **2** illustrates a sectional diagrammatic interior view of an example downdraft table **200** that incorporates a scissor mechanism clamping assembly **220** for an associated filter compartment **224**. Air including suspended particulates, or "dirty air," is shown moving from a perforated work surface **202** and a perforated rear wall **204** into the interior of the downdraft table **200** to a filter insert **210** using a variety of arrows. Filtered air or "clean air" is shown exiting the filter insert **210**, drawn through a fan **248**, and exhausted from the downdraft table **200** via an exhaust port **222** using additional arrows. This general movement of air through the downdraft table **200** is described in detail below.

[0026] The dirty air is generated in an environment above the perforated work surface **202** (horizontal surface) and in the vicinity of an operator of the downdraft table **200**. This air may be harmful to the operator's respiratory health and is thus drawn through holes in the perforated work surface **202**, away from the operator, and into the downdraft table **200**, as illustrated by downwardly directed solid arrows (e.g., solid downward arrow **228**). The air is further drawn through holes in the perforated rear wall **204** (vertical surface), also away from the operator, and into the downdraft table **200**, as illustrated by randomly directed solid arrows (e.g., arrow **240**).

[0027] The dirty air is pulled both downward and rearward through the perforated holes into the downdraft table **200** by the fan **248**, which is located below the perforated work surface **202**. A static vacuum supplied by the fan **248** on the perforated work surface **202** and the perforated rear

wall **204** provides for an even dispersion of airflow over the work surface **202** and the perforated rear wall **204** into the downdraft table **200**. In some implementations, the perforated rear wall **204** is omitted from the downdraft table **200** and all the airflow flows through the perforated work surface **202** into the downdraft table **200**.

[0028] The dirty air is further directed downward into a dirty air intake plenum **242**, as illustrated by downwardly directed broken arrows (e.g., downward broken arrow **244**), and to the left of within the dirty air intake plenum **242**, as illustrated by leftward directed broken arrows (e.g., leftward broken arrow **246**). In other implementations, airflow can be moved in another direction (e.g., rightward, frontward, or backward) within the dirty air intake plenum **242**. The dirty air moves in the leftward direction within the dirty air intake plenum **242** and then downward into a vertical chute **254**, as illustrated by arrows **250**, **252**. In other implementations, there may be other configurations of compartments, chutes, and pathways.

[0029] The dirty air is precluded from reverse airflow through the perforated work surface **202** and/or the perforated rear wall **204** and blow-back towards the operator by a one-way self-closing louvre **230** that allows dirty air to move through the downdraft table **200** in only one direction, thus protecting the operator from potential blow-back of particulate-entrained air. The one-way self-closing louvre **230** is located at the top of the vertical chute **254**. In other implementations, the one-way self-closing louvre **230** is omitted if the risk of blow-back is considered low or negligible. The dirty air moves through a spark arrestor **226** located at one side of the vertical chute **254** between the filter insert **210** and the one-way self-closing louvre **230**, as illustrated by rightward directed solid arrows (e.g., rightward solid arrow **256**). The spark arrestor **226** adds protection against applications that may create sparks or increased risk of a fire such as welding, grinding metal, or plasma cutting.

[0030] The bottom of the vertical chute **254** may include removable clean-out trays **212**, **214**. Particular matter that falls out of the dirty air flow, particularly as it passes through the spark arrestor **226** falls into the removable clean-out trays **212**, **214**. In other implementations, the downdraft table **100** may have greater or fewer removable clean-out trays, and the removable clean-out trays may be located differently than depicted in FIG. 2. For example, a first tray may be adjacent a bottom of the spark arrestor **226** and a second tray may be adjacent a bottom of the filter insert **210** to collect particulate matter. Removal of the clean-out trays does not require removal of the filter insert **210** or any other disassembly of the unit beyond the removal of an access door (not shown, see e.g., access door **116** of FIG. 1).

[0031] The dirty air moves downward through the one-way self-closing louvre **230**, rightward through the spark arrestor **226**, and then toward a center of the downdraft table **200** into a filter compartment **224** that houses a rectangular filter insert **210**. As air moves through the filter insert **210**, particulates in the air attach to the filter insert **210**. Once the dirty air permeates the filter, “clean” air is drawn into a fan plenum **232** and subsequently fan inlet **258**, as illustrated by arrow **260**. An exhaust port **222** is located in the side panel of the downdraft table **200**, and air exits from the fan **248** out of the downdraft table **200** at the exhaust port **222**, as illustrated by arrow **262**. In other implementations, the exhaust port **222** may be vented out a different panel of the downdraft table **200** (e.g., a rear or bottom panel).

[0032] The configuration of the filter insert **210** in the downdraft table **200** is designed to address applications that produce large amounts of dust or fumes on a continuous basis. An operator can observe a pressure gauge (not shown, see e.g., pressure gauge **106** of FIG. 1) mounted on an exterior panel of the downdraft table **200** facing the operator to help determine the cleanliness of the filter insert **210**. The pressure gauge measures the differential pressure across the filter insert **210**. A differential pressure greater than a predetermined differential pressure threshold value indicates that air is being blocked by the dirty filter insert **210** and that it should be cleaned or replaced. When the filter insert **210** has met the predetermined differential pressure threshold and needs to be cleaned or replaced, a notification system can send a signal to alert the operator. The

threshold can be a specific pressure range indicating sufficient particulates have accumulated on the filter to render the airflow insufficient for efficient operation of the downdraft table **200**.

[0033] The filter insert **210** is removable and replaceable by hand by manipulating the scissor mechanism clamping assembly **220**. More specifically, the filter insert **210** is held in place by the scissor mechanism clamping assembly **220** that applies a uniform force on a perimeter of the filter insert **210** to press and secure the filter insert **210** against filter seat **264**. The scissor mechanism clamping assembly **220** functions by transforming rotational motion applied to a scissor mechanism handle (not shown, see e.g., scissor mechanism handle **582** of FIG. 5) in a first direction by an operator into a translational motion that seats the filter insert **210** to the fan plenum **232**. Rotational motion applied to the scissor mechanism handle in a second opposite direction by the operator drives an opposite translational motion that unseats the filter insert **210** from the fan plenum **232** so that the filter insert **210** may be removed from the filter compartment **224**. In some implementations, a filter gasket may form part of the filter seat **264** and may be compressed between the filter insert **210** and the fan plenum **232** to create the desired seal. As a result, changing filter inserts requires no tools and access to the filter compartment **224** requires only the straightforward removal of the access door.

[0034] FIG. 3 illustrates an exploded view of an example filter compartment **324** including a scissor mechanism clamping assembly **320** and an exhaust fan assembly **348**. In various implementations, the filter compartment **324** is a component of a downdraft table, such as downdraft tables **100** and **200** of FIGS. 1 and 2, respectively. The downdraft table functions to draw air including suspended particulates or “dirty air” from a perforated work surface (not shown, see e.g., perforated work surface **202** of FIG. 2) and/or a perforated rear wall (not shown, see e.g., perforated rear wall **204** of FIG. 2) into an interior of the downdraft table to the filter compartment **324**. Filtered air or “clean air” exits the filter compartment **324**, is drawn through the fan assembly **348**, and exhausted from the downdraft table via an exhaust port (not shown, see e.g., exhaust port **222** of FIG. 2). As such, the filter compartment **324** functions to clean the airflow and the fan assembly **348** functions to draw the airflow through the filter compartment **324**. In other implementations, the disclosed technology may be included in an air filtration system independent of a downdraft table.

[0035] The filter compartment **324** is generally a rectangular compartment that is sized and shaped to fit within an internal space of the downdraft table between a dirty air intake plenum, a vertical chute, and/or a spark arrestor (not shown, see e.g., dirty air intake plenum **242**, vertical chute **254**, and spark arrestor **226** of FIG. 2), depending on the specific arrangement of the downdraft table, and the depicted fan assembly **348**. The filter compartment **324** encloses the scissor mechanism clamping assembly **320** and a filter insert (not shown, see e.g., filter insert **210** of FIG. 2). An intake **366** of the filter compartment **324** is generally open to the incoming dirty airflow, while an outlet **368** is mostly closed by fan plenum cover **370**.

[0036] The fan plenum cover **370**, when attached to the outlet **368** of the filter compartment **324** forms a fan plenum (not shown, see e.g., fan plenum **232** of FIG. 2) for directing clean air output from the filter insert to a fan inlet **358** to fan housing **372**. Squirrel cage impeller **374** rotates (illustrated by arrow **328**) to draw the airflow through the filter compartment **324** (illustrated by arrow **340**) and output the airflow from the fan assembly **348** (illustrated by arrow **344**). An impeller cover **376** encloses the impeller **374** within the fan housing **372**. As air moves through the filter insert, particulates in the air attach to the filter insert. Once the dirty air permeates the filter, “clean” air is drawn into the fan plenum and subsequently the fan inlet **358**. An exhaust port (not shown, see e.g., exhaust port **222** of FIG. 2) is located in an exterior panel of the downdraft table, and air exits from the fan housing **372** out of the downdraft table at the exhaust port, as illustrated by arrow **340**.

[0037] An access door **316** provides access to the interior of the filter compartment **324**, including the filter insert, the scissor mechanism clamping assembly **320**, and removable clean-out trays (not

shown, see e.g., trays **212**, **214** of FIG. 2, if present). The access door **316** is secured in position with a securing bar **378** and associated fasteners (e.g., fastener **380**). Use of the securing bar **378** and associated fasteners is only one example of securing the access door **316**. A wide variety of door securing mechanisms are contemplated herein.

[0038] The filter insert is removable and replaceable by hand by manipulating the scissor mechanism clamping assembly **320**. More specifically, the filter insert is held in place by the scissor mechanism clamping assembly **320** that applies a uniform force on a perimeter of the filter insert to press and secure the filter insert against a filter seat (not shown, see e.g., filter seat **264** of FIG. 2). The scissor mechanism clamping assembly **320** functions by transforming rotational motion applied to a scissor mechanism handle (not shown, see e.g., scissor mechanism handle **582** of FIG. 5) in a first direction by an operator into a translational motion that seats the filter insert to the fan plenum. Rotational motion applied to the scissor mechanism handle in a second opposite direction by the operator drives an opposite translational motion that unseats the filter insert from the fan plenum so that the filter insert may be removed from the filter compartment **324**. As a result, changing filter inserts requires no tools and access to the filter compartment **324** requires only the straightforward removal of the access door **316**.

[0039] FIG. 4 illustrates a perspective view of an example filter compartment **424** including a scissor mechanism clamping assembly **420**. In various implementations, the filter compartment **424** is a component of a downdraft table, such as downdraft tables **100** and **200** of FIGS. 1 and 2, respectively. The downdraft table functions to draw air including suspended particulates or “dirty air” from a perforated work surface (not shown, see e.g., perforated work surface **202** of FIG. 2) and/or a perforated rear wall (not shown, see e.g., perforated rear wall **204** of FIG. 2) into an interior of the downdraft table to the filter compartment **424**. Filtered air or “clean air” exits the filter compartment **324**, is drawn through a fan assembly (not shown, see e.g., fan assembly **348** of FIG. 3), and exhausted from the downdraft table via an exhaust port (not shown, see e.g., exhaust port **222** of FIG. 2). As such, the filter compartment **424** functions to clean the airflow as it passes through the filter compartment **424**. In other implementations, the disclosed technology may be included in an air filtration system independent of a downdraft table.

[0040] The filter compartment **424** is a generally rectangular sheet metal box that serves as a housing that is sized and shaped to fit within an internal space of the downdraft table between a dirty side including dirty air intake plenum, a vertical chute, and/or a spark arrestor (not shown, see e.g., dirty air intake plenum **242**, vertical chute **254**, and spark arrestor **226** of FIG. 2), depending on the specific arrangement of the downdraft table, and a clean side including the fan assembly. The filter compartment **424** encloses the scissor mechanism clamping assembly **420** and a filter insert (not shown, see e.g., filter insert **210** of FIG. 2). An intake **466** of the filter compartment **424** is generally open to the incoming dirty airflow, while an outlet **468** is mostly closed by a fan plenum cover (not shown, see e.g., fan plenum cover **370** of FIG. 3).

[0041] The filter insert has a length and height that generally matches a length (l) and height (h) of an interior of the filter compartment **424** so that the filter insert slip fits within the filter compartment **424**. The filter insert has a depth that generally matches a depth (d) of the interior of the filter compartment **424** between the scissor mechanism clamping assembly **420** and the outlet **468** of the filter compartment **424** so that the filter insert is easily inserted within or removed from the interior of the filter compartment **424** when the scissor mechanism clamping assembly **420** is retracted. A space within the interior of the filter compartment **424** defined by the foregoing length (l), height (h), and depth (d) may be referred to herein as a filter cavity that receives the filter insert.

[0042] The filter insert is removable and replaceable by hand by manipulating the scissor mechanism clamping assembly **420**. When the scissor mechanism clamping assembly **420** is extended, the depth (d) of the interior of the filter compartment **424** is reduced and the filter insert is held in place by the scissor mechanism clamping assembly **420** that applies a uniform force on a perimeter of the filter insert to press and secure the filter insert against a filter seat (not shown, see

e.g., filter seat **264**). The scissor mechanism clamping assembly **420** functions by transforming rotational motion (illustrated by arrow **428**) applied to a scissor mechanism handle **482** by an operator into a uniform translational motion (illustrated by arrows **440, 444, 446**). Rotation of the scissor mechanism handle **482** in a first direction by the operator extends the scissor mechanism clamping assembly **420** uniformly to seat the filter insert to the fan plenum. Rotation of the scissor mechanism handle **482** in a second opposite direction by the operator drives an opposite translational motion that unseats the filter insert from the fan plenum so that the filter insert may be removed from the filter compartment **424** and a new filter insert may be placed within the filter compartment **424**.

[0043] FIG. 5 illustrates a perspective view of an example scissor mechanism clamping assembly **520** for a filter compartment (not shown, see e.g., filter compartment **424** of FIG. 4). The filter compartment functions to clean airflow as it passes through the filter compartment. The disclosed technology may be included in a downdraft table, such as downdraft tables **100** and **200** of FIGS. 1 and 2, respectively, or an air filtration system independent of a downdraft table.

[0044] The filter compartment encloses the scissor mechanism clamping assembly **520** and a filter insert (not shown, see e.g., filter insert **210** of FIG. 2). An intake of the filter compartment is generally open to an incoming dirty airflow, while an outlet of the filter compartment discharges a clean airflow. Air flows through the filter compartment from the intake to the outlet and transversely through the scissor mechanism clamping assembly **520** within the filter compartment, as generally illustrated by arrows **544, 546**.

[0045] The scissor mechanism clamping assembly **520** is generally made up of four scissor mechanisms **584, 586, 588, 590**, each of which is generally located at one of the four corners of the scissor mechanism clamping assembly **520**. When installed within a filter compartment, this similarly locates the four scissor mechanisms **584, 586, 588, 590** near the four corners of the filter compartment to distribute force applied by the scissor mechanisms **584, 586, 588, 590** on a filter insert. The scissor mechanisms **584, 586, 588, 590** each includes a pair of interconnected, folding legs that resemble the shape of a pair of scissors. The scissor mechanisms **584, 586, 588, 590** function as mechanical linkages that are used to create extension or retraction from fixed references.

[0046] The scissor mechanism clamping assembly **520** further includes a pair of fixed bars **592, 594** and a pair of translating bars **596, 598**. The fixed bars **592, 594** are secured in fixed positions to an interior of the filter compartment. As the scissor mechanism clamping assembly **520** is extended and retracted, the fixed bars **592, 594** remain in their fixed positions. The translating bars **596, 598** move with the scissor mechanisms **584, 586, 588, 590** as the scissor mechanisms **584, 586, 588, 590** are extended and retracted.

[0047] A pair of spacer bars **583, 585** connect pivot points of scissor mechanisms **584, 588** and scissor mechanisms **586, 590**, respectively. The spacer bars **583, 585** define spacing between the scissor mechanisms **584, 586** and the scissor mechanisms **588, 590** and maintain that spacing as the scissor mechanisms **584, 586, 588, 590** are extended and retracted. The spacer bars **583, 585** further serve selectively move the pivot points of scissor mechanisms **584, 588** and scissor mechanisms **586, 590** closer and further, which drives retraction and extension of the translating bars **596, 598**, respectively.

[0048] An adjuster bar **587** is positioned between the spacer bars **583, 585** and is bolted to spacer bar **583** with bolt **589** and an associated nut (not shown). The adjuster bar **587** is generally smaller than a space between the spacer bars **583, 585** so that rotation of an adjustment screw **591** connecting the adjuster bar **587** to the spacer bar **585** may selectively draw the adjuster bar **587** toward the spacer bar **585** or push the adjuster bar **587** away from the spacer bar **585**, with a variable length of the adjustment screw **591** exposed between the adjuster bar **587** and the spacer bar **585**. Rotation of the adjustment screw **591** changes the spacing between the spacer bars **583, 585**, and consequently the position of folding legs making up each of the scissor mechanisms **584,**

586, 588, 590. This drive changes in extension of the scissor mechanisms **584, 586, 588, 590**, further drives a change in position of the translating bars **596, 598**.

[0049] The filter insert is removable and replaceable from the filter compartment by manipulating the scissor mechanism clamping assembly **520**, primarily by rotating a scissor mechanism handle **582** that drives a corresponding rotation of the adjustment screw **591**. When the scissor mechanism clamping assembly **520** is in an extended position, the filter insert is held in place by the scissor mechanism clamping assembly **520** applies a uniform force on a perimeter of the filter insert to press and secure the filter insert against a filter seat (not shown, see e.g., filter seat **264** of FIG. 2). As detailed above, the scissor mechanism clamping assembly **520** functions by transforming rotational motion applied to the scissor mechanism handle **582** by an operator into a uniform translational motion of the translating bars **596, 598**. Rotation of the scissor mechanism handle **582** in a first direction by the operator extends the translating bars **596, 598** uniformly to seat the filter insert to the fan plenum. Rotation of the scissor mechanism handle **582** in a second opposite direction by the operator drives an opposite translational motion of the translating bars **596, 598** that unseats the filter insert from the fan plenum so that the filter insert may be removed from the filter compartment and a new filter insert may be placed within the filter compartment.

[0050] FIG. 6 illustrates a perspective view of an example scissor mechanism **690** for a scissor mechanism clamping assembly (not shown, see e.g., scissor mechanism clamping assembly **520** of FIG. 5). The scissor mechanism clamping assembly is generally made up of at least two scissor mechanisms, such as scissor mechanism **690**, each of which are generally located on opposing sides or near opposing corners of the scissor mechanism clamping assembly. The scissor mechanism **690** may be similar to any or all of scissor mechanisms **584, 586, 588, 590** of FIG. 5, for example. In various implementations, the disclosed technology may be included in a filter compartment, such as filter compartment **424**; a downdraft table, such as downdraft tables **100** and **200** of FIGS. 1 and 2, respectively; and/or an air filtration system.

[0051] The scissor mechanism **690** includes a pair of interconnected, folding legs **650, 652** that resemble the shape of a pair of scissors. The scissor mechanism **690** functions as a mechanical linkage that is used to create extension or retraction from a fixed reference. The scissor mechanism **690** is attached to a fixed bar **692** and a translating bar **696** (both illustrated in part in FIG. 6). The fixed bar **692** is secured in a fixed position to an interior of the filter compartment or other fixed structure. As the scissor mechanism **690** is extended and retracted, the fixed bar **692** remains in its fixed position, while the translating bar **696** moves with the scissor mechanism **690** as the scissor mechanism **690** is extended and retracted.

[0052] The folding legs **650, 652** are pivotally connected to one another at center pivot **654**. A first end of each of the folding legs **650, 652** is further pivotally connected to one of the fixed bar **692** and the translating bar **696** at fixed pivots **656, 658**, respectively. A second opposite end of each of the folding legs **650, 652** is further slidably and pivotally connected to one of the fixed bar **692** and the translating bar **696** at sliding pivots **660, 662**, respectively. The center pivot **654** and the fixed pivots **656, 658** are assembled as fasteners, such as a machine screw and nut, extending through aligned holes in the folding legs **650, 652**, fixed bar **692**, and the translating bar **696**. The sliding pivots **660, 662** are also assembled as fasteners, such as a machine screw and nut, extending through aligned holes in the folding legs **650, 652** and aligned slots in the fixed bar **692** and the translating bar **696** to enable the sliding connection in addition to the pivoting connection.

[0053] In operation, a separate mechanism (not shown, see e.g., FIG. 5) drives a change in position of the center pivot **654**, as illustrated by arrow **644**. This movement drives a change the position of the folding legs **650, 652** making up the scissor mechanism **690**. This movement further drives a change in extension of the scissor mechanism **690** and a change in position of the translating bar **696**, as illustrated by arrow **646**.

[0054] FIG. 7A illustrates a schematic view of a scissor mechanism clamping assembly **720** for a filter compartment (not shown, see e.g., filter compartment **424** of FIG. 4) in a retracted

orientation. The filter compartment functions to clean airflow as it passes through the filter compartment. The disclosed technology may be included in a downdraft table, such as downdraft tables **100** and **200** of FIGS. **1** and **2**, respectively, or an air filtration system independent of a downdraft table.

[0055] The filter compartment encloses the scissor mechanism clamping assembly **720** and a filter insert (not shown, see e.g., filter insert **210** of FIG. **2**). An intake of the filter compartment is generally open to an incoming dirty airflow, while an outlet of the filter compartment discharges a clean airflow. Air flows through the filter compartment from the intake to the outlet and transversely through the scissor mechanism clamping assembly **720** within the filter compartment, as generally illustrated by arrows **744**, **746**.

[0056] The scissor mechanism clamping assembly **720** is generally made up of four scissor mechanisms, though only two scissor mechanisms **784**, **786** are illustrated in FIGS. **7A** and **7B**. When installed within a filter compartment, each of the four scissor mechanisms are located near one of four corners of the filter compartment to distribute force applied by the scissor mechanisms on a filter insert. While scissor mechanisms **784**, **786** are described in detail below, a similar description could apply to the remaining two scissor mechanisms that are not illustrated in FIGS. **7A** and **7B**.

[0057] The scissor mechanism **784** includes a pair of interconnected, folding legs **750**, **752** that resemble the shape of a pair of scissors. The scissor mechanism **786** also includes a similar pair of interconnected, folding legs **764**, **766**. The scissor mechanisms **784**, **786** function as a mechanical linkages that are used to create extension or retraction from a fixed reference. The scissor mechanisms **784**, **786** are attached to a fixed bar **792** and a translating bar **796**. The fixed bar **792** is secured in a fixed position to an interior of the filter compartment or other fixed structure. As the scissor mechanisms **784**, **786** are extended and retracted, the fixed bar **692** remains in its fixed position, while the translating bar **796** moves with the scissor mechanisms **784**, **786** as the scissor mechanisms **784**, **786** are extended and retracted.

[0058] The folding legs **750**, **752** are pivotally connected to one another at center pivot **754**. A first end of each of the folding legs **750**, **752** is further pivotally connected to one of the fixed bar **792** and the translating bar **796** at fixed pivots **756**, **758**, respectively. A second opposite end of each of the folding legs **750**, **752** is further slidably and pivotally connected to one of the fixed bar **792** and the translating bar **796** at sliding pivots **760**, **762**, respectively. The folding legs **764**, **766** are pivotally connected to one another at center pivot **770**. A first end of each of the folding legs **764**, **766** is further pivotally connected to one of the fixed bar **792** and the translating bar **796** at fixed pivots **772**, **774**, respectively. A second opposite end of each of the folding legs **764**, **766** is further slidably and pivotally connected to one of the fixed bar **792** and the translating bar **796** at sliding pivots **776**, **778**, respectively.

[0059] The center pivots **754**, **770** and the fixed pivots **756**, **758**, **772**, **774** are assembled as fasteners, such as a machine screw and nut, extending through aligned holes in the folding legs **750**, **752**, **764**, **766**, the fixed bar **692**, and the translating bar **696**. The sliding pivots **760**, **762**, **776**, **778** are also assembled as fasteners, such as a machine screw and nut, extending through aligned holes in the folding legs **750**, **752**, **764**, **766** and aligned slots in the fixed bar **792** and the translating bar **796** to enable the sliding connection in addition to the pivoting connection.

[0060] FIG. **7B** illustrates a schematic view of the scissor mechanism clamping assembly **720** of FIG. **7A** in an extended orientation. In operation, a separate mechanism (not shown, see e.g., FIG. **5**) drives a change in position of the center pivots **754**, **770**, as illustrated by arrows **730**, **732**. This movement drives a change the position of the folding legs **750**, **752** making up the scissor mechanism **784**, as illustrated by arrows **734**, **736**. This movement further drives a change the position of the folding legs **764**, **766** making up the scissor mechanisms **786**, as illustrated by arrows **738**, **740**. This movement further drives a change in extension of the scissor mechanisms **784**, **786** and a change in position of the translating bar **796** with reference to the fixed bar **792**, as

illustrated by arrows **724, 726**.

[0061] FIG. **8** illustrates example air filtration operations in an example downdraft table. In various implementations, the downdraft table includes an integrated filter compartment. Other implementations omit the downdraft table and integrate the filter compartment within another structure. Operations **805, 810, 815, 820, 825** are directed to using a scissor mechanism clamping assembly for filter replacement within the filter compartment.

[0062] Opening operation **805** opens an access door to permit an operator access to the filter compartment. In various implementations, the access door is opened, but remains attached to the downdraft table or access door is completely moved from the downdraft table for the opening operation **805**. Retracting operation **810** retracts the clamping assembly from a dirty filter insert within the filter insert. The clamping assembly includes at least two scissor mechanisms, each mounted on opposing sides of an inlet to the filter compartment. One end of each of the scissor mechanisms is mounted to a fixed position within the inlet. The clamping assembly further includes two translating bars. An opposite end of each of the scissor mechanisms is mounted to one of the translating pressure bars. The clamping assembly still further includes an adjustment screw to change the position of folding legs making up each of the scissor mechanisms and thereby change the position of the translating bars. The operator may rotate the adjustment screw in a first direction using a scissor mechanism handle to drive the retraction of the clamping assembly. In implementations where no filter insert is present, the retracting operation **810** provides clearance for insertion of a replacement filter insert in operation **815**, discussed below.

[0063] Replacing operation **815** replaces the dirty filter insert with a clean filter insert within the filter compartment. Specifically, the operator may physically remove the dirty filter insert from the filter compartment by sliding the dirty filter insert form between the clamping assembly and a fan plenum and out of the open access door. The operator may then physically insert the clean filter insert into the filter compartment by sliding the clean filter insert through the open access door and between the clamping assembly and the fan plenum. An extending operation **820** extends the clamping assembly to press the clean filter insert against a filter seat at the fan plenum. The operator may rotate the adjustment screw in a second direction using the scissor mechanism handle to drive the extension of the clamping assembly. A closing operation **825** replaces the access door, if removed, and closes the access door after the extending operation **820**. This secures the filter compartment.

[0064] A drawing operation **830** draws dirty air through one or more of a perforated work surface and a perforated rear wall of the downdraft table and into the filter compartment using a fan assembly. The fan assembly provides static vacuum in the downdraft table to provide an even dispersion of airflow through the perforated work surface and/or the perforated rear wall. Thus, the dirty air can be pulled downward, rearward, and/or horizontally through the perforated holes into the downdraft table by the fan assembly, which is located within the downdraft table behind the filter insert. In other implementations, for example in an air filtration system, air can be pulled through an intake, panel, nozzle, or other entry way.

[0065] An arresting operation **835** arresting sparks using a spark arrestor located between the filter insert and the perforated work surface and/or the perforated rear wall. The spark arrestor arrests sparks as air is drawn through the downdraft table. The spark arrestor adds protection against applications that may create sparks or increased risk of a fire such as welding, grinding metal, or plasma cutting. Dirty air may move through the spark arrestor downward in the downdraft table, and then move in a direction toward the center of the downdraft table into the filter compartment that houses the filter insert.

[0066] A collecting operation **840** collects particulate matter that separates from the dirty air flow prior to filtering operation **845**, discussed below, in one or more removable clean-out trays. The operator may periodically remove the clean-out trays to empty the particulate matter. A filtering operation **845** filters the dirty air within the filter compartment to produce clean air output from the

filter compartment. As air moves through the filter insert, particulates in the air attach to the filter insert, thereby cleaning the air flow.

[0067] An exhausting operation **850** exhausts the clean air out of the downdraft table. In some implementations, an exhaust port is located in the back of the downdraft table, and air exits out a back panel of the downdraft table. In other implementations, the exhaust port may be vented out a different panel of the downdraft table (e.g., a bottom panel). The downdraft table may include a self-closing louvre. The self-closing louvre forces the exhausting operation **850** to flow air in a singular direction, thereby precluding particulate from blow-back through the perforated work surface and/or the perforated rear wall.

[0068] The logical operations making up the implementations described herein are referred to variously as a method, operations, steps, objects, or modules. Furthermore, it should be understood that logical operations may be performed in any order, adding or omitting operations as desired, unless explicitly claimed otherwise or a specific order is inherently necessitated by the claim language.

[0069] The structure and/or functionality of the downdraft tables, filter compartments, and/or scissor mechanism clamping assemblies may be different than that illustrated in FIGS. **1-8** and described herein. For example, the scissor mechanism clamping assemblies may be integrated in structures beyond downdraft tables or used in combination with other systems for air filtration. Similarly, the arrangement of the components within the downdraft tables and/or filter compartments are provided for purposes of illustration and not of limitation, and some components and/or interconnections may be omitted for purposes of clarity. It will be further appreciated that the downdraft tables, filter compartments, and/or scissor mechanism clamping assemblies may not include all of the components shown in FIGS. **1-8**, may include other components that are not explicitly shown in FIGS. **1-8**, or may utilize an architecture completely different than that shown in FIGS. **1-8**.

[0070] The above specification, examples, and data provide a complete description of the structure and use of example implementations of the invention. Since many implementations of the invention can be made without departing from the spirit and scope of the invention, the invention resides in the claims hereinafter appended. Furthermore, structural features of the different implementations may be combined in yet another implementation without departing from the recited claims. The implementations described above, and other implementations are within the scope of the following claims.

Claims

1. A downdraft table comprising: a perforated work surface; a clamping assembly including: two scissor mechanisms, each mounted on opposing sides of a filter intake, one end of each of the scissor mechanisms mounted to a fixed position within the filter intake; two translating bars, an opposite end of each of the scissor mechanisms mounted to one of the translating bars; an adjustment screw to change the position of folding legs making up each of the scissor mechanisms and thereby change the position of the translating bars; and a filter cavity to receive a filter insert, the filter insert to be pressed against a filter seat at a fan plenum by the translating bars; and a fan assembly to draw air through the perforated work surface, the clamping assembly, the filter insert, and the fan plenum and exhaust filtered air out of the downdraft table.
2. The downdraft table of claim 1, further comprising: a perforated rear wall, wherein the fan assembly is further to draw air through the perforated wall.
3. The downdraft table of claim 1, wherein the clamping assembly further includes: two additional scissor mechanisms, one end of each of the additional scissor mechanisms mounted to a fixed position within the filter intake, wherein: the scissor mechanisms and the additional scissor mechanisms are mounted near opposing corners of the filter intake; an opposite end of each of the

- additional scissor mechanisms are mounted to one of the translating bars; and the adjustment screw further to change the position of folding legs making up each of the additional scissor mechanisms.
4. The downdraft table of claim 1, further comprising: an access door to provide an operator access to the filter insert.
 5. The downdraft table of claim 1, wherein the filter seat includes a compressible gasket.
 6. The downdraft table of claim 1, further comprising: a removable clean-out tray adjacent to the filter insert for particulate collection.
 7. The downdraft table of claim 1, further comprising: a spark arrestor oriented between the perforated work surface and the filter insert, wherein the fan assembly is further to draw air through the spark arrestor prior to the filter insert.
 8. The downdraft table of claim 7, further comprising: a removable clean-out tray adjacent to the spark arrestor for particulate collection.
 9. The downdraft table of claim 1, further comprising: a one-way, self-closing louvre to prevent reverse airflow through the perforated work surface.
 10. A filter compartment comprising: a housing including an filter intake and a filter outlet; a clamping assembly including: two scissor mechanisms, each mounted on opposing sides of the filter intake, one end of each of the scissor mechanisms mounted to a fixed position within the filter intake; two translating bars, an opposite end of each of the scissor mechanisms mounted to one of the translating bars; an adjustment screw to change the position of folding legs making up each of the scissor mechanisms and thereby change the position of the translating bars; and a filter cavity to slidably receive a filter insert, the filter insert to be compressed against a filter seat at the outlet by the translating bars.
 11. The filter compartment of claim 10, wherein the clamping assembly further includes: two additional scissor mechanisms, one end of each of the additional scissor mechanisms mounted to a fixed position within the filter intake, wherein: the scissor mechanisms and the additional scissor mechanisms are mounted near opposing corners of the filter intake; an opposite end of each of the additional scissor mechanisms are mounted to one of the translating bars; and the adjustment screw further to change the position of folding legs making up each of the additional scissor mechanisms.
 12. The filter compartment of claim 10, further comprising: an access door to selectively provide access to the filter cavity to permit replacement of the filter insert.
 13. The filter compartment of claim 10, wherein the filter seat includes a compressible gasket.
 14. The filter compartment of claim 10, further comprising: a removable clean-out tray adjacent to the filter insert for particulate collection.
 15. A method of using a clamping assembly for filter replacement within a filter compartment comprising: retracting the clamping assembly from a dirty filter insert, wherein the clamping assembly includes: two scissor mechanisms, each mounted on opposing sides of an inlet, one end of each of the scissor mechanisms mounted to a fixed position within the inlet; two translating bars, an opposite end of each of the scissor mechanisms mounted to one of the translating bars; and an adjustment screw to change the position of folding legs making up each of the scissor mechanisms and thereby change the position of the translating bars, wherein rotation of the adjustment screw in a first direction drives retraction of the clamping assembly; replacing the dirty filter insert with a clean filter insert within the filter compartment; and extending the clamping assembly to press the clean filter insert against a filter seat at a fan plenum.
 16. The method of claim 15, wherein the filter compartment is within a downdraft table, the method further comprising: drawing dirty air through one or more of a perforated work surface and a perforated rear wall of the downdraft table and into the filter compartment using a fan assembly; filtering the dirty air within the filter compartment to produce clean air output from the filter compartment; and exhausting the clean air out of the downdraft table.
 17. The method of claim 15, further comprising: opening an access door to the filter compartment prior to the retracting operation; and closing the access door after the extending operation.

- 18.** The method of claim 16, further comprising: collecting particulate matter that separates from the dirty air prior to the filtering operation in one or more removable clean-out trays.
- 19.** The method of claim 16, further comprising: precluding particulate from blow-back through the perforated work surface or the perforated rear wall with a one-way self-closing louvre.
- 20.** The method of claim 16, further comprising: arresting sparks using a spark arrestor located between the filter insert and the perforated work surface or the perforated rear wall.
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